Reg. No. | | | | | | | | | | |

[12ME82]

EIGHTH SEMESTER B.E. EXAMINATION, JUNE - 2016

POWER PLANT ENGINEERING

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

1.	a.	List the methods of out-plant handling of coal.	03
	b.	Explain functioning of screw conveyer with a sketch. List its advantages and disadvantages.	09
	c.	Comment briefly on ash handling systems.	02
2.	a.	Illustrate classification of combustion system used for coal burning.	03
	b.	Schematically narrate working of a chain grate stoker.	06
	c.	Share the importance of a cyclone burner highlighting its merits	05
3.	a.	List all the essential components of diesel electric plant and mention their role.	04
	b.	Explain double circuit cooling system with reference to its necessity in diesel electric power plants.	06
	c.	With a sketch explain lubrication system having continuous centrifugal cleaning in a diesel plant.	04
4.	a.	Explain functioning of a gas turbine power plant with the aid of a diagram.	06
	b.	Classify and compare different gas turbine power plants. Justify with the aid of sketches.	08
5.	a.	Elaborate on site selection parameters for a hydro-electric power plant.	06
	b.	Depict general arrangements of storage type hydro-electric project and explain its operation.	08
6.	a.	With the aid of schematic representation, narrate the working principle of horizontal axis two blade windmill.	06
	b.	What is the specialization of vertical axis wind mill? Describe its constructional features as compared to horizontal axis windmill.	06
	c.	Give the detailed classification of tidal power plants.	02
7.	a.	Label all the general components of nuclear reactor.	04
	b.	How are nuclear reactors classified? List all the basis is detail.	04
	c.	Comment briefly on all the factors which control the site selection for nuclear power plant.	06

8. a. The cost of electrical energy depends on fixed cost and running cost. Elaborate on their constituent cost. **04**

b. Following data pertains to a power plant of 120MW capacity. **10**

The capital cost	- Rs. 1500/KW
Interest of depreciation	- 10% on capital
Annual running charges	- Rs. 20×10^6
Profit to be gamed	- 10% of capital
Energy consumed by auxiliaries	- 5% of generated
Annual load factor	- 0.6
Annual capacity factor	- 0.5

Calculate : i) reserve capacity and ii) cost of generation per kWh